

JWALIN SHAH

Fremont, CA · (408) 908-9659 · jwalinshah13@gmail.com · github.com/jwalin-shah · linkedin.com/in/jwalin-shah
AI Systems Engineer — agent reliability · evaluation · local inference

PROFESSIONAL EXPERIENCE

Break the Web | AI Systems Engineer — LiveLM

May 2026 – Present

- Architecting LiveLM end-to-end so models answer from **current** world knowledge: local retrieval over ~80K indexed facts, REST/OpenAPI + MCP tool surfaces, and ok / partial / stale / miss routing that forces fallback instead of fluent hallucination (~11 ms warm p50).
- Built the measurement loop that makes that architecture honest — 71 hand-labeled questions × 3 serving paths, 512 story-anchored probes, and sidecar checks on retrieval, freshness, answerability, and abstention.
- Shipping the ChatGPT Custom GPT path against live HybridSearch and tuning serving behavior from real benchmark and probe failures — close the loop between product surface and eval signal.

tinyhumansai | Core Open-Source Contributor — OpenHuman

Jan 2026 – Present

- Own the personal-memory / ingest problem in OpenHuman's Rust core: prototype and harden on-device life capture so the agent reasons over **your** messages, mail, calendar, and contacts — not a generic chat log.
- Built hybrid retrieval for that index (FTS5 + sqlite-vec) and staged ingest pipelines across iMessage, Gmail, Calendar, Contacts, and chronicle streams — correctness of ingest before embed.
- Shipped contact identity resolution and scoring via CNContactStore (cross-platform, mockable source); carved ~1,500 lines of CLI adapters into domain-owned Rust modules with 196 tests across MCP, services, utilities, and autocomplete.
- Hardened the agent's production edges around that memory work: memory citations and namespaces, typed controller/RPC registries, privacy and security routes, and chat-stream reliability so personal context doesn't leak or flake under real use.

Skild AI | Data Operations Lead — ML Systems & Reliability

San Mateo, CA · Jul–Nov 2025

- Built the data validation, collection, and evaluation spine across five robotic platforms — Python/Apps Script automation cut ops overhead ~50% and lifted downstream task success ~40%.
- Ran technical execution for 25+ live Series C demos with zero failures; scaled a 30+ operator program with scheduling, QA, training, compliance, and runbooks that held under investor pressure.

SELECTED SYSTEMS

Bridge — Reliable Coding-Agent Execution

Go · Seatbelt · Worktrees · Formal Methods

- Spawn → verify → deliver for coding agents: deny-default macOS sandboxing, leased worktrees, allowlisted adapters, and **independent** fresh-checkout verification — the worker never certifies its own success.
- Made reliability checkable: 77 pipeline invariants, 52 checked or proved, 28 property-based checkers, 8 Z3 proofs, 5 TLA+ models (incl. 1.58M-state exploration), and Lean proofs.

OfficeQA Arena — Grounded Reasoning Evaluation

Python · Retrieval · Tool Use

- Hit 184.5/246 (75%) at \$1.71 after 12 architecture rounds and ~4,400 evaluations — then proved the bottleneck: evidence selection caused 48% of failures; correctly grounded Python answers had **0%** arithmetic errors.

Voice Engine — On-Device Speech Infrastructure

Swift · CoreML · MLX

- Shipped fully local dictation (Moonshine + VAD + punctuation + native injection). Across 26 ASR configs and 560 clips, moonshine-tiny matched canary-qwen-2.5B accuracy (0.2177 vs 0.2162 WER) at **14× lower RTF** (0.05 vs 0.73) — real-time on device, no network.
- 348/349 tests pass; one CoreML OS regression called out honestly. Harness splits failures into transcription, punctuation, vocabulary, and runtime so model picks are evidence, not vibes.

EDUCATION

University of Texas at Dallas — B.S. Computer Science, Dec 2024

Ultimate Frisbee — Captain · UTD Chess Team — member